

RANI LAXMI BAI MEMORIAL SCHOOL



SESSION 2023-24

PHYSICS PROJECT

TOPIC:- p-n JUNCTION DIODE

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CLASS-XII SEC-A2

BOARD ROLL NO.-

BONAFIDE CERTIFICATE

This is to certify that PALAK KUMARI SINGH of class XII A2, a bonafide Student at a Rani Laxmi Bai Memorial School, C-block Indira Nagar, Lucknow has successfully completed her project on a subject of Physics and found the same under the curriculum of Central Board Of Secondary Education, Delhi

Signature of internal examiner:

Signature of external examiner:

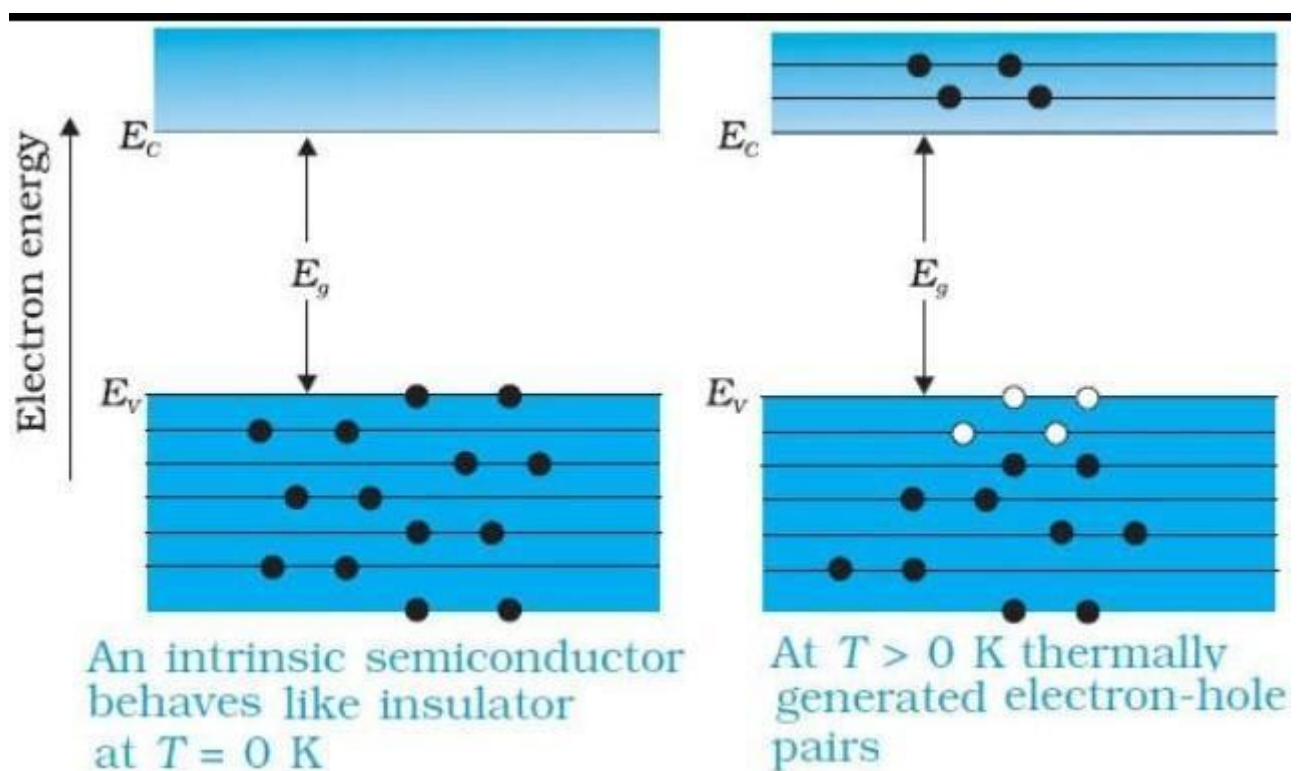
Signature of Principal: _____

ACKNOWLEDGEMENT

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SEMICONDUCTOR

Semiconductors possess resistivity and conductivity intermediate to metals and conductors.

Properties:

1. Semiconductors have much higher resistivity than metals.
2. These have a temperature coefficient of resistivity(α) that is both negative and high, that is, the resistivity of semiconductors decreases rapidly with temperature while that of metals increases.
3. These have a considerable lower number density(n) of charge carriers (charge carriers per unit volume) than metals.

Classification Of Semiconductors:

A. **Intrinsic Semiconductors**: The pure semiconductors in which the electrical conductivity is totally governed by the electrons excited from the valence band to the conduction band and in which no impurity atoms are added to increase their conductivity are called intrinsic semiconductors and their conductivity is called intrinsic conductivity. Electrical conduction in pure semiconductors occurs by the means of electron-hole pairs. In an intrinsic semiconductor,

$$n_e = n_h = n_i$$

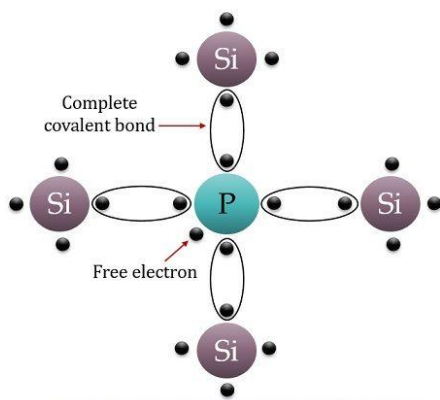
where,

n_e = the free electron density in the conduction band

n_h = the hole density in the conduction band.

n_i = the intrinsic carrier concentration.

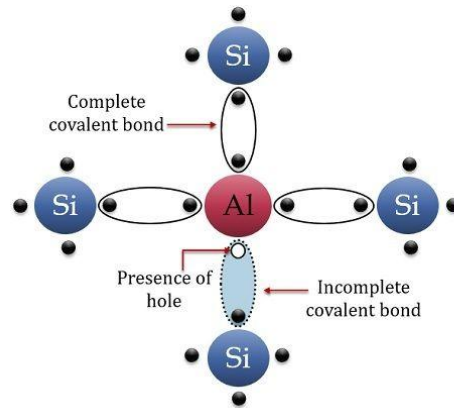
+



- Si = Intrinsic semiconductor atom
- P = Pentavalent impurity atom

Formation of N type extrinsic semiconductor

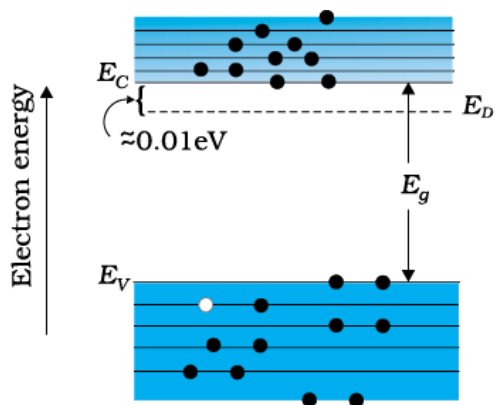
Electronics Desk



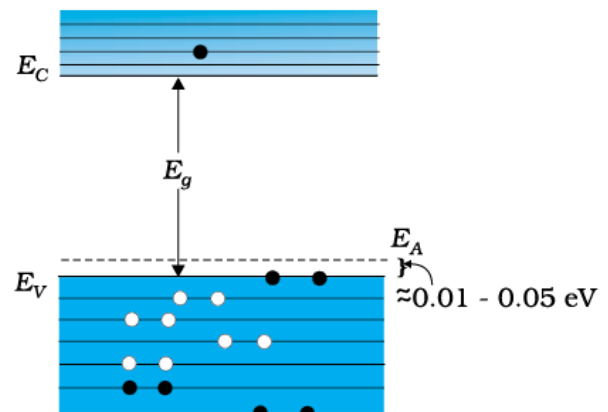
- Si = Intrinsic semiconductor atom
- Al = Trivalent impurity atom

Formation of P type extrinsic semiconductor

Electronics Desk



(a) $T > 0\text{K}$
one thermally generated electron-hole pair + 9 electrons from donor atoms



(b) $T > 0\text{K}$

B. **Extrinsic conductors:** Semiconductor doped with suitable impurity atoms so as to increase its conductivity is called an extrinsic semiconductor.

Extrinsic semiconductors are of two types,

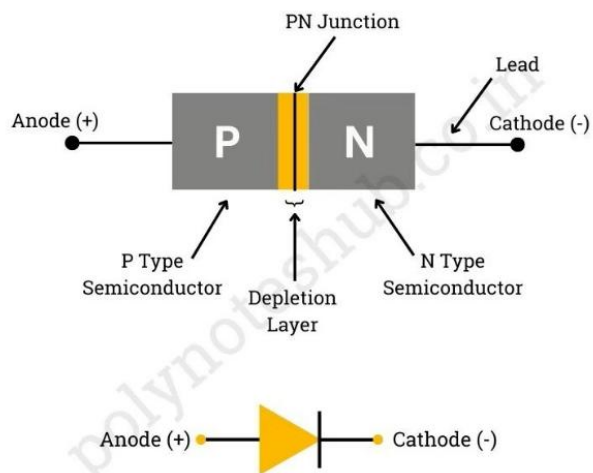
- i. n-type semiconductor
- ii. p-type semiconductor

- **n-type semiconductor:** The pentavalent impurity atoms are called the donors because they donate electrons to the host crystal and the semiconductor doped with donors is called n-type semiconductor. In n-type semiconductors, electrons are the major charge carriers and holes are minority charge carriers. Thus,

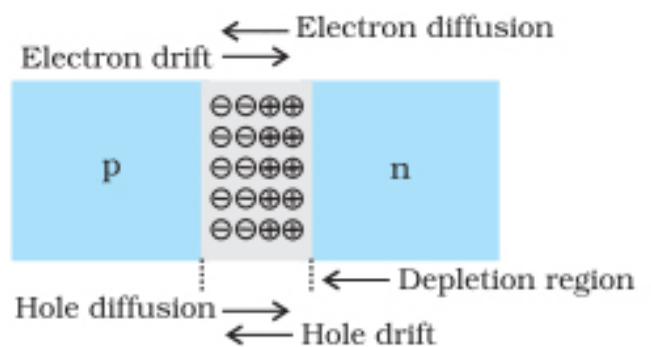
$$n_e = n_o > n_h$$

- **p-type semiconductor:** trivalent The impurity atoms are called acceptors because they create holes which can accept electrons from the nearby bonds. A semiconductor doped with acceptor type impurities is called a p-type semiconductor. In p-type semiconductor, holes are the majority charge carriers and electrons are minority charge carriers. Thus,

$$n_h = n_o > n_e$$



Construction of PN Junction Diode and Symbol



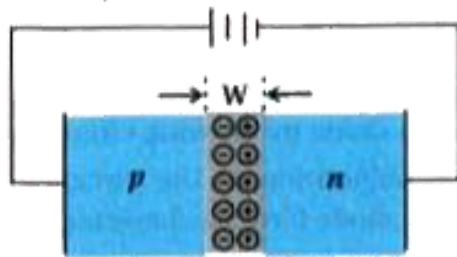
p-n junction formation process.

FORMATION OF P-N JUNCTION DIODE

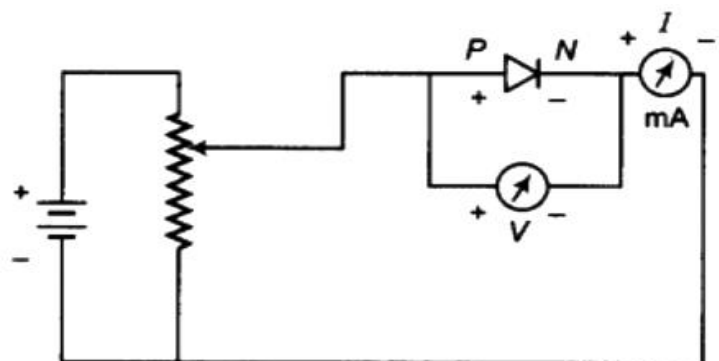
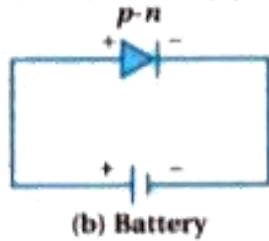
Two important processes occur during the formation of p-n junction: diffusion and drift. We know that in an n-type semiconductor, the concentration of electrons (no of electrons per unit volume) is more compared to the concentration of holes. Similarly, in p-type semiconductors, the concentration of holes is more than the concentration of electrons. During the formation of p-n junction and due to the concentration gradient across p and n side, holes diffuse from p to n side and electrons diffuse from n to p side. This motion of charge carriers gives rise to diffusion across the junction.

When an electron moves from n→p it leaves behind an ionised donor on n side. This ionized donor (positive charge) is immobile as it is bonded to the surrounding atoms. As the electrons continue to diffuse from n→p side a layer of positive charge (or positive space-charge region) on n side of junction is developed.

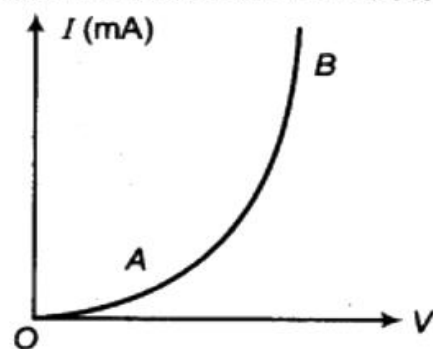
Similarly, when a hole diffuses from p→n side due to the concentration gradient, it leaves behind an ionized acceptor (-ve charge), which is immobile. As holes continue to diffuse, a layer of negative charge (or negative space-charge region), of the p-side of the junction is developed. The space charge region developed on either side of the junction together is known as depletion region. Initially diffusion current is large and drift current is small. As the diffusion current is large and process continues, the space charge regions on either side of junction extend, thus, increasing the strength and drift current. This process continues until the diffusion current equals the drift current. Thus a p-n junction diode is formed



(a) *p-n* junction diode under forward bias
The figure for the circuit (b) is as follow.



Experimental arrangement for P-N junction forward bias characteristic



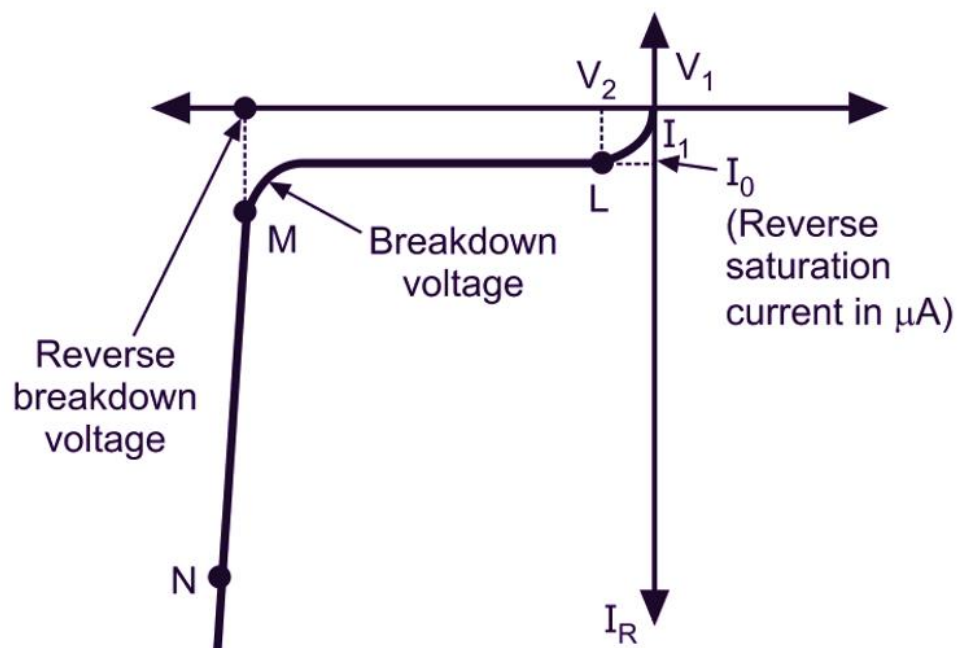
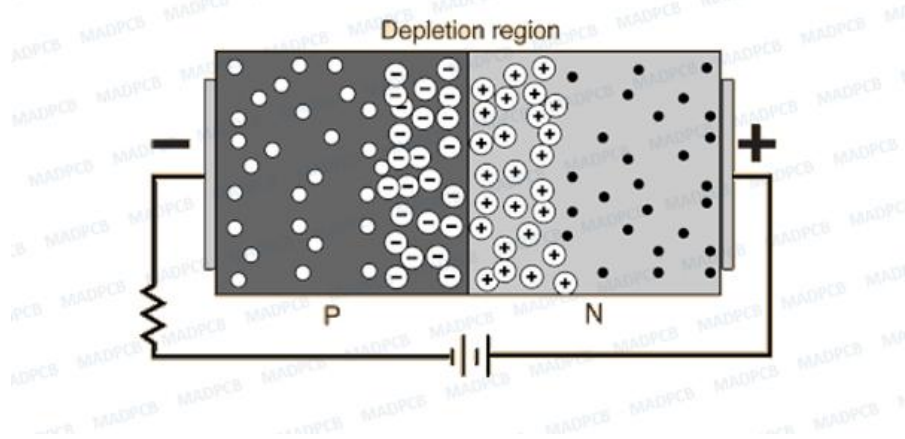
P-N junction forward bias characteristic curve

FORWARD BIASING

If the positive terminal of a battery is connected to the p-side and the negative terminal to the n-side, the p-n junction diode is said to be forward biased. The applied voltage V opposes the barrier voltage V_B . As a result,

- i. The effective barrier potential decreases to $V_B - V$ and hence the energy barrier across the junction decreases.
- ii. The majority charge carriers, that is, holes from p side and electrons from n-side begin to flow toward junction.
- iii. Electrons and holes diffuse into the depletion region.
- iv. The effective resistance across the p-n junction decreases.

Reverse Bias (PN Junction)



REVERSE BIASING

If the positive terminal of a battery is connected to the n-side and the negative terminal to the p-side, then the p-n junction is said to be reverse biased. As shown in the figure, the applied voltage V and the barrier potential V_B are in the same direction, as a result,:

- i. The effective barrier potential decreases to $V_B + V$ and hence the energy barrier across the junction increases.
- ii. The majority charge carriers move away from the junction, increasing the width of the depletion region.
- iii. The resistance of the p-n junction becomes very large.
- iv. No current flows across the junction due to majority charge carriers.

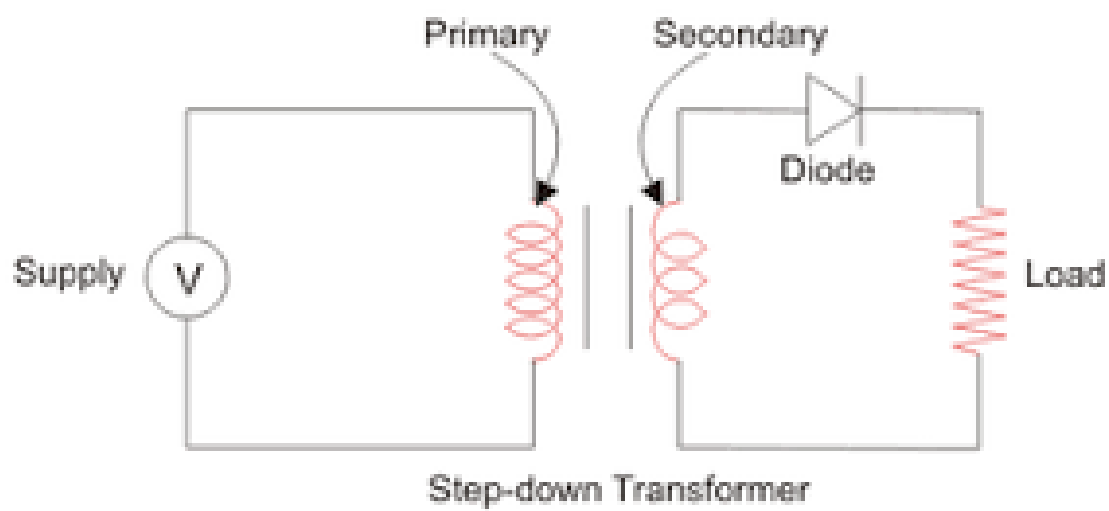
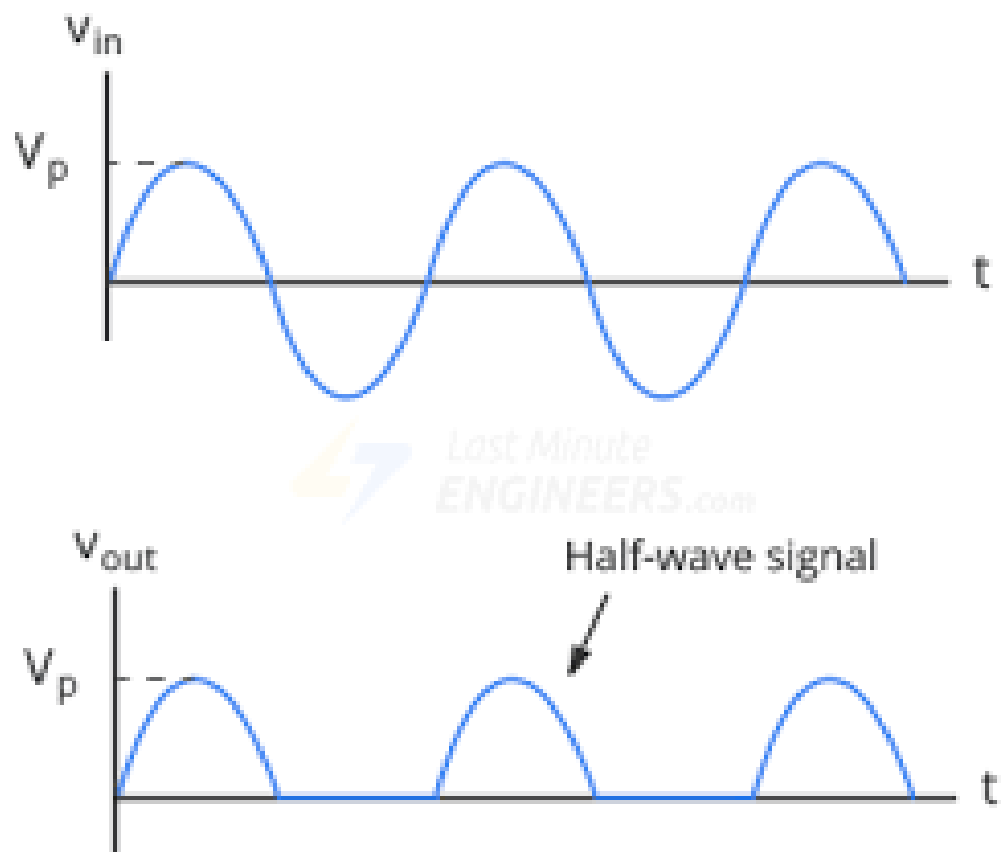


Figure - 2



JUNCTION DIODE AS A RECTIFIER

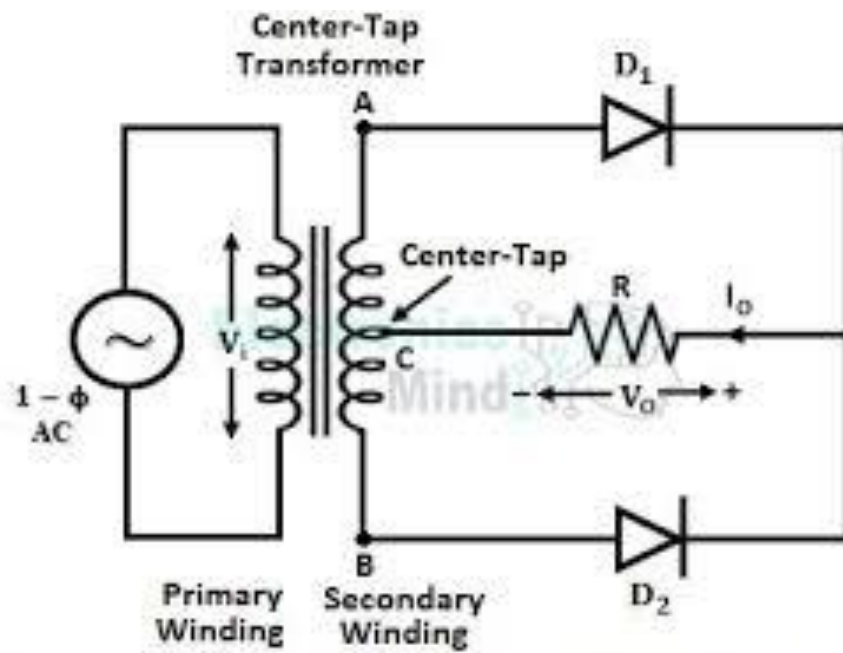
Rectifier: The process of converting alternating current into direct current is called rectification and the device used for this is called a rectifier.

The p-n junction can be used as:

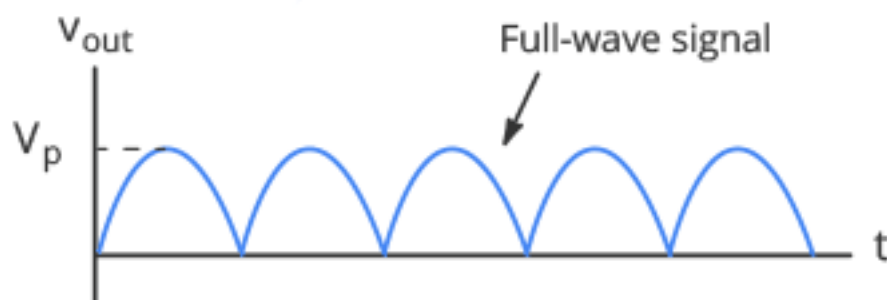
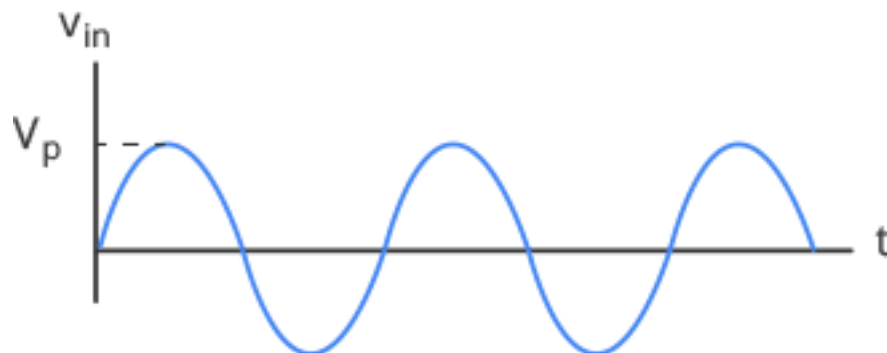
1. Half Wave Rectifier: A Rectifier which rectifies each half of one A.C input supply cycle.

Principle: It is based on the principle that junction diode of a low resistance path when forward biased, a high resistance path when reverse biased. It gets forward biased during only half cycle and reverse biased during the next half-cycle. Thus, the output obtained is alternate half cycles if an A.C input source.

Working: During the positive half cycle of A.C the end a is +ve and the end B is -ve, that is, and A is at the higher potential with respect to B, the diode D gets forward biased, it offers low resistance and a current I flows through R_1 . The upper end of R_1 will be at positive potential with respect to the lower end. D.C output voltage can be obtained across the lower resistance R . During second half cycle of A.C input voltage, the end A is -ve and the end B is +ve. The diode gets reverse biased, it offers higher resistance and the current I does not flow through the lower resistance R_1 . No D.C output can be obtained across lower resistance R_L . at The magnitude of D.C output sources across R_i during the first cycle the time will be proportional to the magnitude of the current through R_L that is proportional to the no of majority charge carriers crossing the junction which ultimately depends upon the value of A.C input at that time hence during the first half cycle of A.C input voltage D.C. output varies.



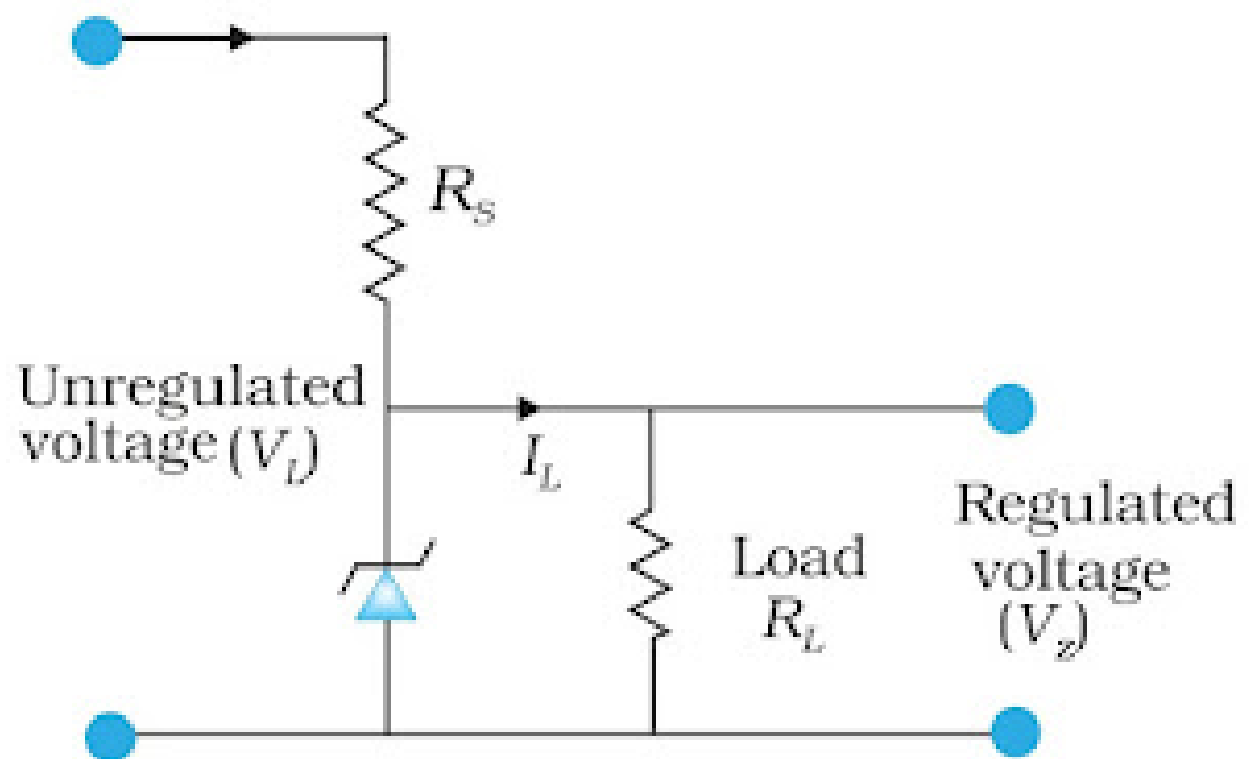
Center-tapped Full-wave Rectifier with R-load



2. **A Full Wave Rectifier:** a rectifier which rectifies both halves of each A.C input cycle is called full wave rectifier.

Principle: It works on the principle that a junction diode offers low resistance during forward biasing and high resistance when reverse biasing. Therefore, the two diodes are connected in such a manner that is one diode gets forward biased during the 1st half-cycle of AC input, the other gets reverse biased, but when the next opposite half cycle comes, the former diode gets reverse biased while the latter gets forward biased. Thus the output is obtained during both half-cycles of A.C input.

Working: During the positive half-cycle of A.C input, the end A is positive and the end B is negative, with respect to center tap T, then the Diode D_1 gets forward biased and conducts currents along the path DXYTA. The diode D_2 is reverse biased and does not conduct. During the negative half-cycle of AC input, the end A becomes negation and the end B becomes positive, with respect to the center tap T. The diode D_2 gets forward biased and conducts current along the path BD₂XYTB. As during both half-cycles of AC input the current through load R_L flows in the same direction X to Y, so we get a pulsating DC voltage. Since the output voltage across the load resistance R_L is obtained for both half-cycles of input AC, this process is called full wave rectification and the arrangement used is called full wave rectifier.

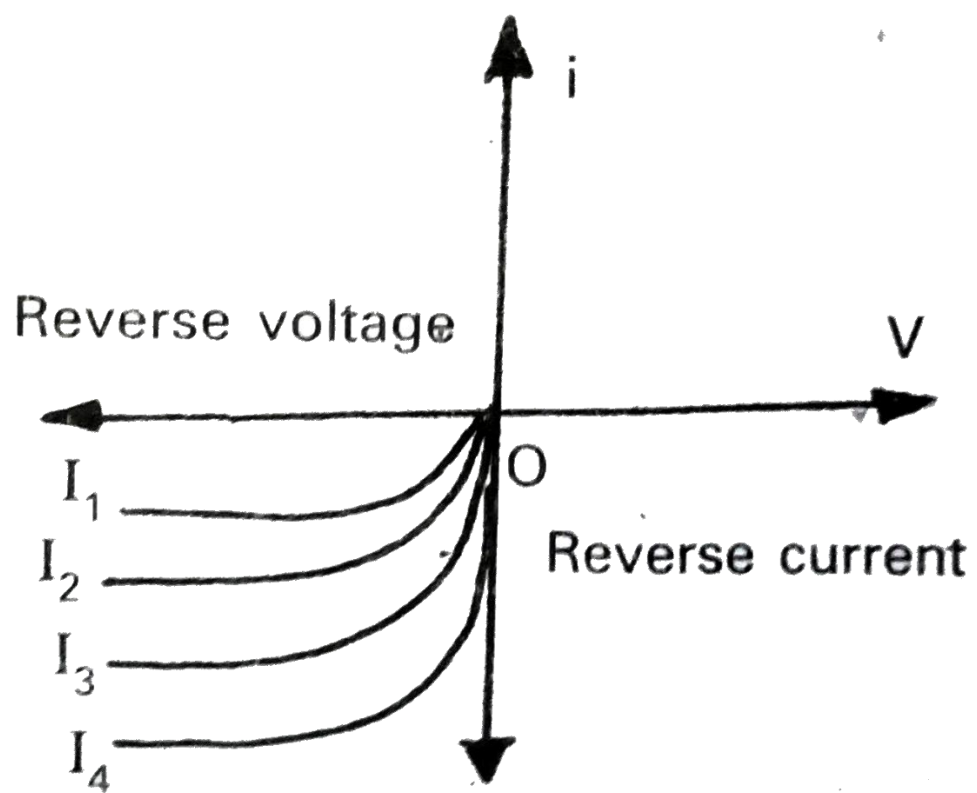
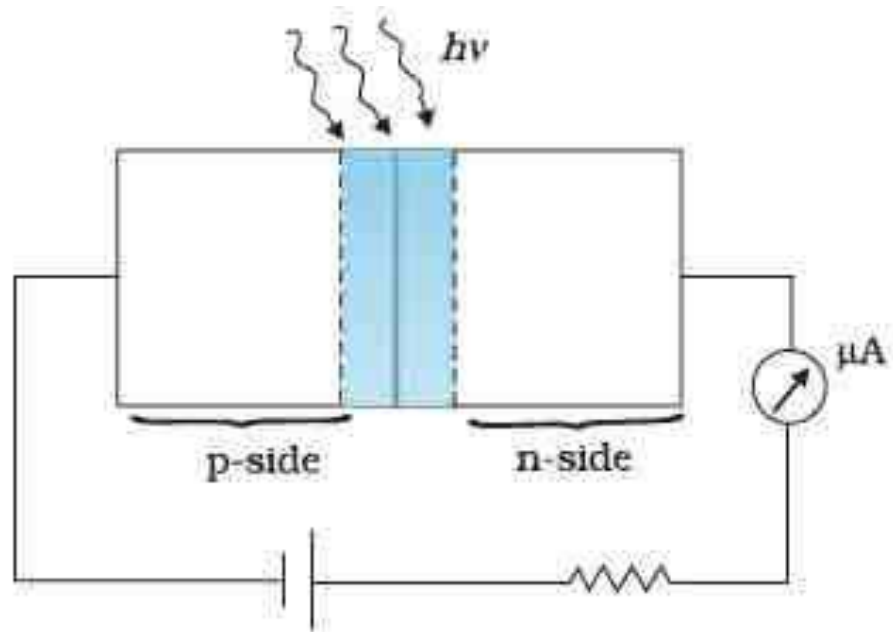


ZENER DIODE

A junction diode specially designed to operate only in the reverse break down region continuously (without getting damaged) is called Zener Diode.

Principle: When a Zener Diode is operated in the reverse breakdown region, the voltage across it remains practically the constant (equal to the breakdown voltage V_Z) for a large change in the reverse current.

Working: The Zener Diode is connected in reverse bias to a source of fluctuating D.C. through dropping R_s . The voltage gets divided between R and Zener Diode. The output is obtained across the load resistor R_L connected in parallel with Zener Diode. If the input voltage increases, the current through R_s and Zener diode also increase. This increase the voltage drop across the Zener Diode. This is because in the breakdown region, Zener Voltage remains constant even though the current through the Zener Diode changes. Similarly, if the input voltage decreases, the voltage across R , decreases without any change in the voltage across the Zener Diode. Thus any increase/ decrease of input voltage results in increase/ decrease across Zener Diode. Hence, the Zener Diode acts as a regulator.



PHOTONIC P-N JUNCTION DEVICE

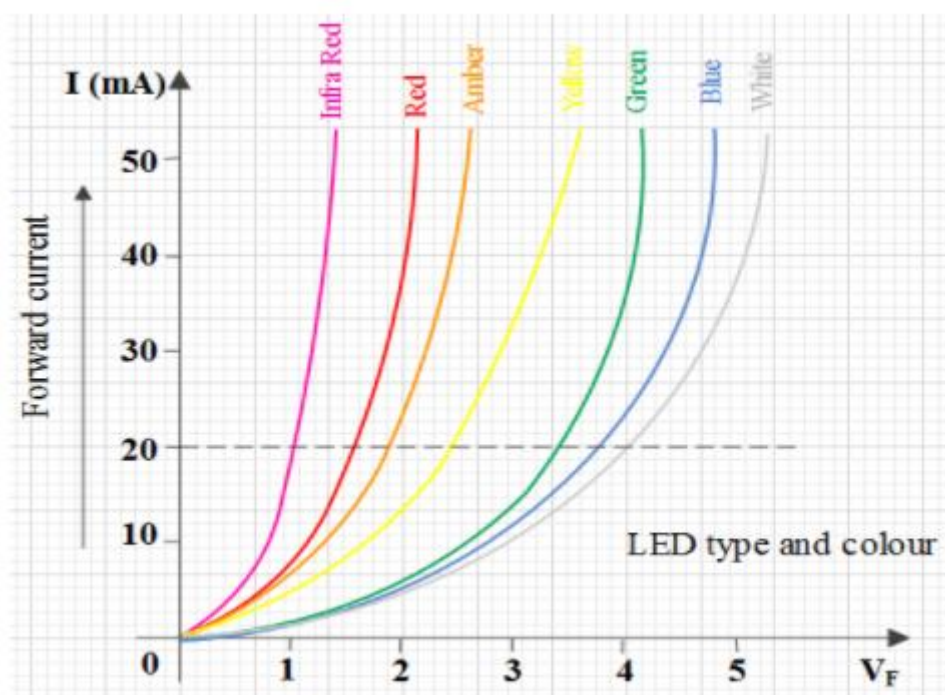
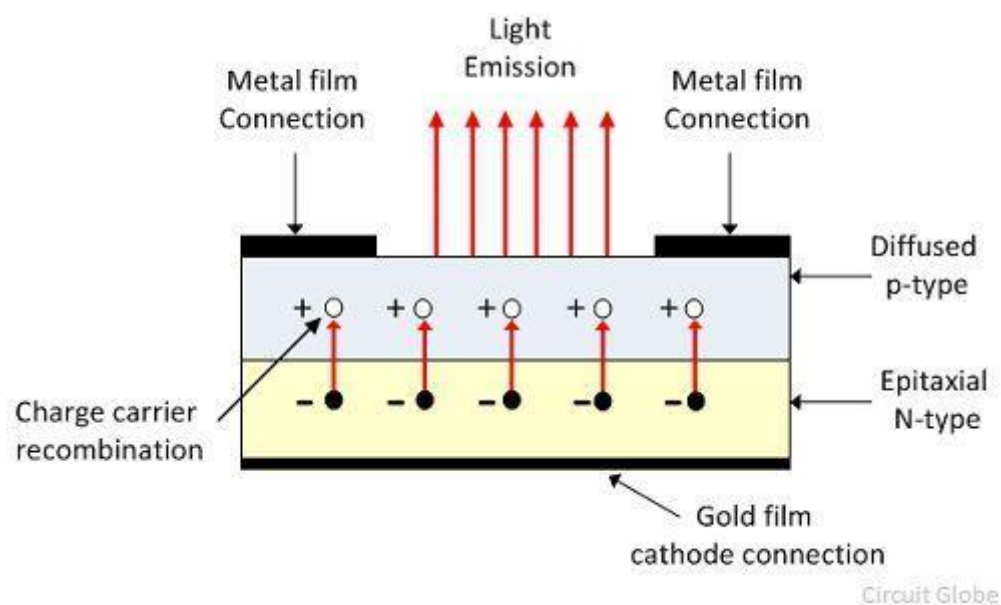
These devices are designed so that the current through them changes either by causing excitation (from valence band to conduction band) by light photons, or conversely through electron excitation by a suitable bias voltage resulting in the emission of light photons. These semiconductors are called photonic or opto- electronic device

Photodiode: A photo diode is a p-n junction fabricated from a photosensitive semiconductor and provided with a transparent window so as to allow light to fall on its junction.

Working: A photodiode is subjected to photons in the form of light which affects the generation of electron-hole pairs. If the energy of the falling photons ($h\nu$) is greater than the energy gap (E) of the semiconductor material, electron-hole pairs are created near the depletion region of the diode. The electron-hole pairs created are separated from each other before recombining due to the electric field of the junction. The direction of the electric field in the diode forces the electrons to p side. As a result of the increase in the number of electrons on the n-side and holes on the p-side, a rise in the electromotive force is observed. Now when an external load is connected to the system, a current flow is observed through it. The more the electromotive force created, the greater the current flow. The magnitude of the electromotive force created depends directly upon the intensity of the incident light. This effect of the proportional change in photocurrent with the change in light intensity can be easily observed by applying a reverse bias. Since photodiodes generate current flow directly depending upon the light intensity received, they can be used as photo detectors to detect optical signals. Built-in lenses and optical filters may be used to enhance the power and productivity of a photodiode.

Uses of photodiode:

- Photodiodes are used in solar panel cells.
- Photodiodes are used in logic circuits.



1. Light Emitting Diode: It is a heavily forward biased p-n junction spontaneously converts the biasing electrical energy to optical energy.

Working: When the diode is forward biased, the minority electrons are sent from p→n while the minority holes are sent from n p. At the junction boundary, the concentration of minority carriers increases. The excess minority carriers at the junction recombine with the majority charges carriers. On recombination, the energy is released in the form of photons with energy $h\nu > E_g$. When the forward bias of the diode is small, the intensity of light is small. As the forward current increases the intensity of light increases and reaches a maximum. Further increases in the forward current decrease the light intensity. LEDs are biased such that light emitting efficiency is maximum.

Two important features of LED:

- The colour of light emitted by an LED depends on its band-gap energy.
- The intensity of light emitted is determined by the forward current conducted by the p-n junction.

Choice of the Semiconductor material used: Visible LEDs are available for variety colours. Energy range of the visible region is 1.8eV to 2.8eV. hence, those materials with energy gaps in this range can be used. Si and Ge cannot be used in LED, because their energy gaps are less than 1.8eV. Gallium arsenide-phosphide is used for making LEDs of different colours.

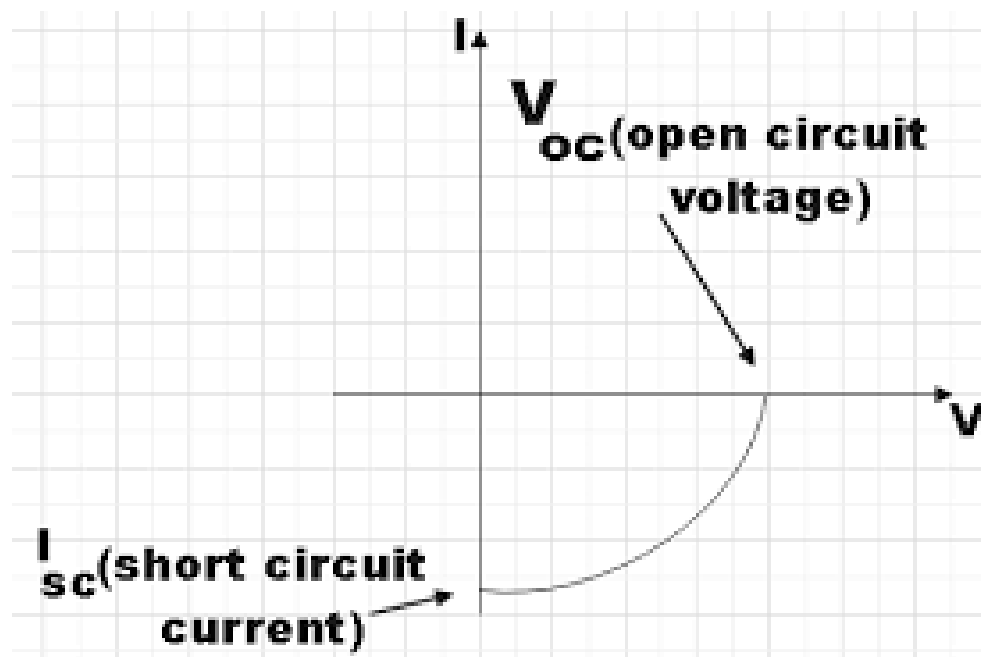
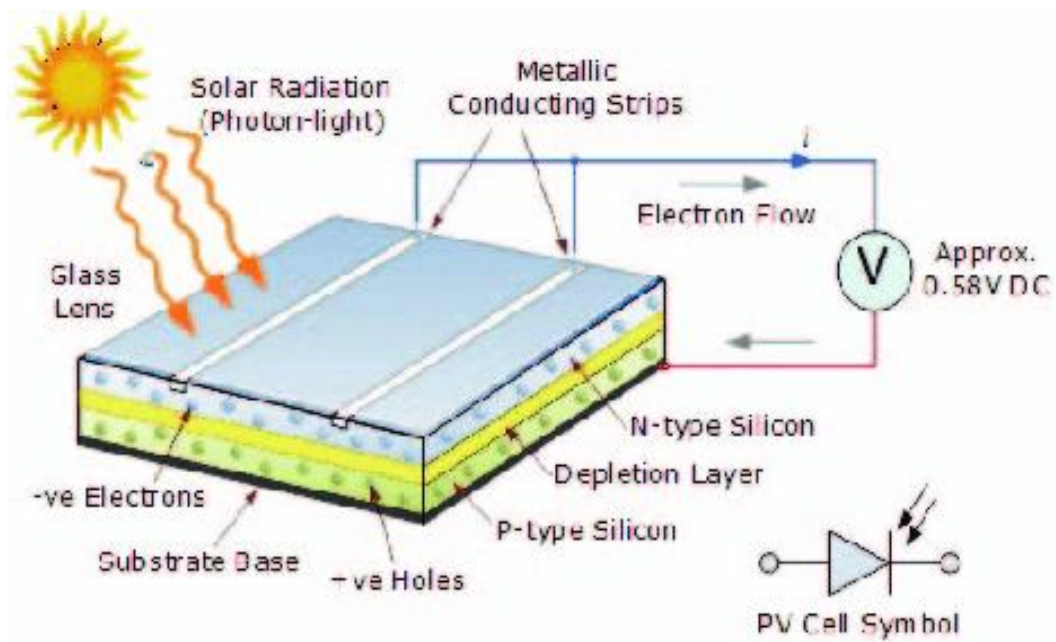
Gallium phosphide → red or green, Gallium arsenide phosphide → red or yellow → red or green, Gallium arsenide phosphide → red or yellow Gallium arsenide → IR or red, Gallium nitride → blue .

Advantage of LED over conditional Incandescent bulb:

- LEDs consume less power, and they require low operational voltage.
- No warm-up time is needed for LEDs.
- The emitted light is monochromatic.
- They exhibit long life and ruggedness.

Uses of LED: LEDs find applications in various fields, including optical communication, alarm and security systems, remote-controlled operations, robotics, etc. It finds usage in many areas because of its long-lasting capability, low power requirements, swift response time, and fast switching capabilities. Below are a few standards LED uses:

- Used for TV back-lighting.
- Used in displays
- Used in Automotives
- LEDs used in the dimming of lights.



2. Solar Cell: It consists of a p-n junction made silicon or gallium arsenide. Here a thin layer of p-type is grown (by diffusion of suitable acceptor impurity or by vapour deposition) on an n-type semiconductor. The top of the p-layer is provided with a few finger electrodes. This leaves an open ending space for light to reach the thin p-layer and hence the underlying p-n junction. It is a junction diode that converts solar energy into electrical energy. Working: The generation of emf by a solar cell, when light falls on, it is due to the following three basic processes: generation, separation and collection- (i) generation of e-h pairs due to light (with $h\nu > E_g$) close to the junction; (ii) separation of electrons and holes due to electric field of the depletion region. Electrons are swept to n-side and holes to p-side; (iii) the electrons reaching the n-side are collected by the front contact and holes reaching p-side are collected by the back contact. Thus p-side becomes positive and n-side becomes negative giving rise to photovoltage. When an external load is connected as shown in the Fig. 14.25(a) a photocurrent I_L flows through the load.

Advantages of solar cell: Reduce the cost of energy bills
Renewable source of energy. Environmentally friendly Low maintenance costs Independent energy .

Uses of solar cell: Charge storage battery during day time and use the battery at night. Used as light meters in photography In spacecrafts to provide electrical energy

BIBLIOGRAPHY

In the successful completion of this project, I have taken help from the following sources:-

- SL Arora class XII Part 1
- NCERT Physics class XII part 1
- electronicshub.com